Meeting Summaries of February 26

Date: February 26
Time: 18:00 - 18:55 PM(PDT)
Attendees: Prof. Jan, Dr. Jabed, Dr. Yue, Jian and Feng
Topic: A 35 minutes overview of the whole computation and the work done so far, and 20 minutes discussion on the potential imporvements and the next steps.

Summary of Discussion

- Feng gave an **overview** of the adapted computation method and the important figures so far, provided some background on where the results can be found and how the results are reproducible and consistent with the algorithms as Jian used.

  But it did **not** include the details of the methods, for example, what are the transformation steps before ordination approaches and what do they ecologically mean, and how do these ordination approaches operate on multiple gradients and fit with the practical relevance of out targets.

- Prof. Jabed suggested Feng to elaborate on the **specific steps in each computation**, they should be elucidated with the **site level of granularity**, and some mathematical details may be needed to explain these operations.

- Feng asked questions about fitting the piecewise quantile regression models that the confidence intervals seem to shift away from the point estimated parameters, and the searching method for breakpoints seems to be incomplete (grid-searching method is not very sufficient).

  Prof. Jabed gave some suggestions on the fitting of the piecewise quantile regression models:

  1. To different parts (quantile levels) of the distribution, there might have different estimates for the thresholds. Locations of breakpoints estimated from different quantile levels can move around on the stress-response plane, that is shifting on the stress-axis and the response-axis.
     When moving the estimation work (extreme quantiles) to the **edges** of the distribution, there may be **less valid samples** for fitting the quantile regression models, so the confidence intervals may be wider and less connected with the point estimates.

     My reflections on CIs of fitting PQRM around edges

     This problem might lay in the **design of optimization problems** for fitting the piecewise quantile regression models, the objective function is to keep around $\tau \times 100$ percent samples below the fitted line and the rest above the fitted line.
     When use the wild-bootstrap method to make synthetic samples for estimating set of parameters and deriving confidence intervals, the edge quantil-distributed synthetic samples might variant a lot from the original samples. The more edging (extreme quantil levels) the synthetic samples are, the more variant they are from the originally edging samples, then leads to the wider and the overly shifted confidence intervals.

- Prof. Jan suggested Feng to talk with Jian about the ecological details and meanings behind the computation methods, and there are **ecological reasons** when talking about the breakpoints in different quantile levels of the model:

  1. When talking about **different quantile levels** of response variable, we are considering the community positions that are relatively better (higher quantile, like 0.8) or worse (lower quantile, like 0.2) than the average position (quantile 0.5) conditional on the same stress level.

     My reflections on ecological reasons behind the breakpoints and quantiles

     When breakpoints of different quantile levels are detected at different stress levels, it is worth and necessary to **connect** the relative community positions **(quantiles)** with the critical **stress levels** (at breakpoints) to understand the ecological implications. For example, to the 0.8 quantile level, the breakpoint detected at a higher stress level means the the community staying at the upper edges (the better community) will **hold higher stress levels** before reaching the breakpoint (structural collapse point).
     But after this this breakpoint, this upper edge community (0.8 quantile) may hold a steeper decline than the average community (quantile 0.5) and its ecological condition would therefore deteriorate faster than the average community.
     Such pollution-driven changes finally lead the sites to **aggregate around** a narrower range of lower ecological condition (response variable) accompnaied by higher stress levels. In the scatter plot of the stress-response relationship, it looks like the sites are getting clustered with smaller variation in the response variable when **moving towards the higher stress levels**.

Action Items

- Feng will talk with Jian about the **ecological reasons and details** behind the computation steps, to complement the ecological relevance of the study.
- Regarding the computational steps, Feng needs to elaborate on the **specific operations** applied on the **site(row in the data) level of granularity**, making sure they are statistically solid.
- The **breakpoints located at different quantile levels** (edges in the distribution) are worth to be **highlighted** and carefully interpreted, because they reveal or stay consistent with an important but easily overlooked ecological patterns.

Follow-Up

Temporary meeting time: March 5, 18:00 - 18:40 PM (Vancouver), which corresponds to 19:00 - 19:40 PM (Calgary) and 20:00 - 20:40 PM (Winnipeg)

Supportive materials

- A meeting recording has been restored on Feng's local drive.